

Particle-in-Cell simulations of lithium redeposition in plasma sheaths under oblique magnetic fields

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R. Avril,^{1,2,a)}  S. Heuraux,²  E. Gravier,²  F. A. Volpe,¹  and J. Moritz² 

AFFILIATIONS

¹Renaissance Fusion, 14 Rue Jean Pierre Timbaud 38600 Fontaine, France

²Université de Lorraine, CNRS, IJL, F-54000 Nancy, France

^{a)}Author to whom correspondence should be addressed: romain.avril@univ-lorraine.fr

ABSTRACT

Compact nuclear fusion reactors will require advanced plasma-facing components capable of withstanding the higher heat and neutron fluxes characteristic of these machines. Liquid metal walls, based for instance on lithium, have emerged as a promising alternative to solid walls. This study investigates lithium redeposition on plasma-facing walls in a deuterium plasma environment using a self-consistent 1D3V Particle-in-Cell code. The model accounts for the interplay between lithium evaporation and the electric potential with a simplified but relevant physics. Four key parameters were analyzed for their impact on lithium redeposition: the ratio γ_s of the Li^+ outflux over the D^+ influx, the magnetic field inclination θ , the ionization mean free path Λ_{iz} , and the magnetic field intensity B . Results indicate that γ_s and θ significantly influence redeposition, leading to redeposition rates higher than 95%, while variations in Λ_{iz} and B had secondary effects. In general, an increase in one of these parameters drives to a lower redeposition, except for θ , where potential “bumps” formed under specific conditions, alters ion trajectories and results in a non-monotonic redeposition dependency. The role of lithium in flattening the potential profile and its evaporation rate highlight the need to take Li impurity into account in the calculation of the potential profile.

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I. INTRODUCTION

Tungsten and other solid plasma-facing materials, when adequately cooled, can cope with the heat fluxes of some experimental fusion reactors. ITER, for example, is expected to produce 500 MW of heat, which will impinge on approximately 630 m^2 of first wall and will result in peak heat fluxes of 15 MW/m^2 on the divertor.^{1,2} In compact commercial fusion reactors, however, smaller surfaces will be exposed to comparable or higher amounts of heat per unit time. For instance, a stellarator concept under investigation at Renaissance Fusion³ envisions thermal powers in excess of 2 GW on $200\text{--}500 \text{ m}^2$ of first wall, resulting in peak heat fluxes possibly higher than 15 MW/m^2 in flux-concentrating regions such as divertors. The corresponding high heat and neutron fluxes would rapidly erode and damage any solid material, calling for its frequent replacement. Liquid metal plasma-facing components (PFC) have emerged as a promising alternative regarding this issue, in particular for compact concepts. Due to their thermodynamic and neutronic properties, candidate materials include lithium (Li) and tin (Sn) with melting temperatures of, respectively, 181 and 232°C .

As a matter of fact, Li, Sn, and Li–Sn alloys have already been tested in both stellarator^{4,5} and tokamak^{6–10} environments over the past years

and have shown a better protection of the vessel compared to solid PFC, both during regular operations and disruption-like events. In the aforementioned experiments, the liquid metal film was thin (millimeters or less) and either static or flowing slowly (cm/s), at a temperature barely above the melting point. In fact, several beneficial effects are already observed for walls coated with solid Li, due to the high chemical reactivity of Li, making it a getter for plasma impurities, thus renewing the interest in low-recycling regimes.^{11,12} Some specific future reactor designs, such as the one developed by Renaissance Fusion, might benefit from full coverage by thicker, faster, hotter liquid walls. Regarding the thickness, thin films can absorb charged particles and low-energy photons, but burning plasmas will produce copious amounts of neutrons emitted in all directions, whose absorption requires actually greater thicknesses.¹³ Removing heat will also require a certain speed that can be attained thanks to electromagnetic forces.³ Finally, liquid walls will act as the first working fluid in the power-plant, and higher temperatures will boost thermodynamic efficiency and facilitate the extraction of lithium tritide by a simple precipitation scheme.^{14,15}

A rich physics is associated with this high expected temperature T_s of the liquid metal surface as temperature-dependent erosion was

observed due to both thermal and adatom evaporation,¹⁶ in addition to the physical sputtering also observed in solids. These liquid-specific erosion mechanisms have motivated both experimental¹⁷ and numerical studies¹⁸ related to the Magnum-PSI linear device. These works primarily focused on contamination, power handling, impurity and fuel retention, and vapor shielding.¹⁹ In designs where the liquid layer is positioned close to the plasma, the estimated maximum working surface temperature approaches 700 °C. However, investigations suggest that specific concepts, such as the vapor box divertor,²⁰ could leverage higher temperatures and associated evaporation rates by confining the vapor cloud further from the plasma. Although these studies have provided key insights into liquid lithium PFCs, further work is still needed to better understand sheath boundary conditions. For example, experimental measurements are currently used as boundary conditions for SOLPS-ITER simulations.¹⁸

According to literature, wall-emitted impurities can be redeposited back to the wall by several complex processes, whose depend on the magnetic configuration, but above all on the plasma conditions (density, temperature, collisionality, etc.) and the impurity energy and ionization position, characterized by dominant atomic reactions and erosion mechanisms.

- If the eroded impurity—under neutral or ionized form—reached regions of the Scrape-Off Layer (SOL) far enough from the sheath and presheath, then it can be redeposited due to an adequate parallel to the magnetic field velocity (only for ions) or due to diffusion or entrainment processes, as it was numerically and experimentally observed in experiments.^{21–23} As magnetic field lines connect two points of the vessel surface, ionized impurities can also be redeposited on another region of the reactor by following the field lines. This last mechanism is correlated to the competition between the parallel and perpendicular transport in the SOL, and depends of the neoclassical and turbulent effects.
- Closer to the wall, in the sheath and presheath region, the redeposition of charged impurities ionized near the surface is generally driven by a combination of Lorentz and electric forces (Fig. 1). The electric force pushes impurities back toward the emitting surface. This force is the strongest within the sheath (dark purple in Fig. 1), whose extent scales with the Debye length, but it decreases as one moves into the quasi-neutral region. In magnetized plasmas, the quasi-neutral region consists of two distinct areas. Far from the surface, within the collisional presheath, ions are gradually accelerated along magnetic field lines to reach the parallel acoustic velocity $c_{s//}$. At that point, they enter the “Chodura” or magnetic pre-sheath (light purple in Fig. 1),²⁴ whose scale is set by the ion Larmor radius. In the latter region, ions are deflected perpendicularly to the wall before entering the sheath at the Bohm velocity c_s .²⁵ The potential drop across these regions is also known to vary with the magnetic field inclination θ relative to the surface and the plasma collisionality.²⁶ Since the electric force is strong in the sheath but generally weaker in the pre-sheaths, the probability of impurity redeposition depends on the precise location of ionization.

For heavy impurities such as tungsten, sputtering from solid PFCs at high initial velocity can produce particles with a gyroradius larger than the ionization mean free path, leading to prompt redeposition. This phenomenon has been extensively investigated through

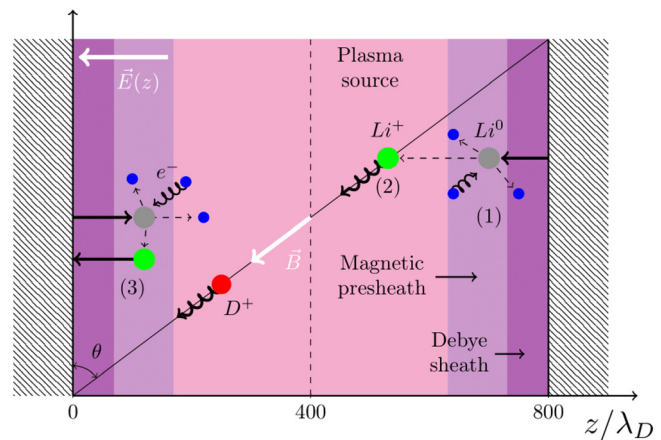


FIG. 1. Schematic representation of three physical processes simulated by the PIC code in a bulk plasma, magnetic presheaths and sheaths comprised between two liquid walls: (1) electron-induced ionization of evaporated neutral atoms, (2) motion of ions (Li^+ and D^+) in the plasma, and (3) redeposition of ionized impurity.

numerical modeling and analytical calculations, showing that redeposition rates can be larger than 90% provided that neutral tungsten atoms are ionized within the sheath.^{27–30} In contrast, for lighter impurities, especially lithium released through high-temperature surface evaporation, the redeposition process in the sheath region is still poorly understood and remains largely unexplored. One of the main differences between tungsten and lithium lies in the erosion process and in the amount of impurities released into the plasma: while the tungsten erosion yield is only a few percents, it can exceed unity for lithium at high temperature. As a result, a substantial population of Li^+ can build up in the sheath and pre-sheath regions, altering both the electric potential profile and the sheath characteristics.^{17,28} A self-consistent treatment is then required to evaluate the Li redeposition rate, as it alters the sheath electric field that returns Li ions to the wall, in contrast to tungsten, for which the trace impurity assumption generally remains valid.

Consequently, the aim of this study is to advance the understanding of sheath physics in liquid lithium PFCs, a field that remains relatively unexplored. Using a Particle-in-Cell code, we investigate sheath and presheath physics, as well as the in-sheath redeposition of ionized impurities originating from evaporation in a collisionless plasma. Here, only ionization is considered, allowing us to draw broader conclusions applicable not only to fusion research but also to other studies involving sheaths influenced by both magnetic fields and evaporated surfaces. This work complements previous PIC-based redeposition estimates,³¹ as well as earlier research that coupled a hybrid fluid-kinetic scrape-off layer (SOL) code with a PFC heat transport code to self-consistently link surface temperature, evaporation flux, and impurity plasma transport.^{32,33} While it is often assumed—much like in Monte Carlo codes such as ERO2.0³⁴ and GTR³⁵—that sheath properties remain unaffected by lithium evaporation, this study demonstrates that this assumption does not always hold.

The remainder of the paper is organized as follows. Section II describes the PIC code and the geometry considered, detailing the Li injection procedure and the plasma parameters used in the simulations. Section III presents the results from various PIC simulations,

including a discussion of sheath formation in the transient regime and the effect of the liquid metal surface temperature (Subsection III A), the influence of the magnetic field inclination relative to the wall on the redeposition process (Subsection III B), and the impact of the ionization mean free path—through variations of the plasma density—and the magnetic field strength on redeposition rates (Subsections III C and III D, respectively).

II. PARTICLE-IN-CELL CODE AND SIMULATION MODEL

A. Presentation of the model

The in-house Particle-in-Cell (PIC) code used in this study is one-dimensional (1D) in space, corresponding to the direction normal to the lithium wall (denoted z), and three-dimensional (3D) in velocity space, in order to account for the gyration of charged particles (Deuterium D^+ and Lithium Li^+ ions, as well as electrons) around magnetic field lines. The simulated plasma is embedded between two conductive liquid walls separated by 800 Debye lengths (λ_D), as depicted in Fig. 1. This configuration was chosen for numerical symmetry, but in the present work we focus on the redeposition process on a single wall. We also consider a magnetic field B intersecting both surfaces at an angle θ with respect to them. The Deuterium plasma is injected into the simulated system at a constant flux Γ_D . Newly born D^+ ions and electrons are positioned randomly around the plasma center ($z = 400\lambda_D$), following a Gaussian distribution in space, with a standard deviation typically set to $200\lambda_D$. The flux Γ_D is chosen of the order of the Bohm flux, calculated as the product of the nominal plasma density n_0 and the ion acoustic speed $c_s = \sqrt{k(T_e + T_D)/m_D}$, where m_D and T_D are the Deuterium mass and temperature, respectively. Therefore, since the incoming lithium flux Γ_{Li} is also imposed (see Sec. II B), the densities of both ion species and electrons evolve so as to satisfy the flux conditions at the walls while preserving ambipolarity. To prevent plasma cooling due to energetic particles losses (mainly fast electrons at the walls), both D^+ ions and electrons velocities in the simulated system are periodically reset according to the initial Maxwellian distribution.

In the code, the time step is determined by electron kinetics and is set to $dt = 2\pi/(10\omega_{ce})$. The length of the simulation box is chosen to be sufficiently large compared to the ionization mean free path, and the spatial step is $dx = \rho_{Le}/4$, where ρ_{Le} is the electron Larmor radius. A Courant test, defined as $Co = (dt\sqrt{T_e/m_e})/dx > 1$, is also performed to ensure that physical information does not traverse more than one spatial step during a given time step. If this condition is not satisfied (which rarely occurs), the time step is adjusted according to $dt = 0.25dx/\sqrt{T_e/m_e}$.

Finally, particle pushing for velocity $\mathbf{v}_p$, charge q_p , and mass m_p is performed using the Boris algorithm to solve Newton's second law (1), while the Poisson equation (2) is solved using a finite element method to self-consistently determine the electric potential ϕ from the density profiles of singly charged particles (electrons, D^+ , and Li^+),

$$\frac{d\mathbf{v}_p}{dt} = \frac{q_p}{m_p} \left(-\frac{d\phi}{dz} \mathbf{e}_z + \mathbf{v}_p \times \mathbf{B} \right), \quad (1)$$

$$\frac{d^2\phi}{dz^2} = \frac{e}{\epsilon_0} (n_e - n_D - n_{Li}), \quad (2)$$

where $\mathbf{e}_z$ is the unit vector associated with the z -direction (see Fig. 1) and n_e , n_D , and n_{Li} are the electron, D^+ , and Li^+ densities, respectively.

This formulation enables a self-consistent treatment that accounts for the influence of lithium ions on the electric potential and, consequently, on their redeposition.

B. Li^+ injection procedure

The uniform and constant temperature of the lithium wall T_s is set between 500 and 700 °C at the beginning of the simulation, to be relevant for specific reactor designs, such as the one proposed by Renaissance Fusion. The 700 °C surface temperature aligns with Renaissance Fusion's reactor design specifications, which considers a Li–LiH binary alloy (fully liquid above 690 °C). For our simplified pure Li analysis, we anticipate that exceeding this temperature may result in excessive evaporation. The incoming lithium flux Γ_{Li} in our model is determined by the surface temperature through the classical Hertz–Knudsen–Langmuir relation,

$$\Gamma_{Li}(T_s) = \frac{P_v(T_s)}{\sqrt{2\pi k T_s m_{Li}}}, \quad (3)$$

where m_{Li} and k are the lithium mass and the Boltzmann constant, respectively, and P_v is the lithium vapor pressure given by the Hick's correlation of the National Institute of Standards and Technology (NIST).³⁶ The trajectories of Li atoms emitted from the wall are ballistic, as they are insensitive to electromagnetic fields. Their initial velocities are sampled from a Maxwellian distribution in the x and y directions [Eq. (4)], and from a Rayleigh distribution [Eq. (5)] in the z direction, perpendicular to the wall, which is always taken as positive,

$$f_x(v_x) = f_y(v_y) = \sqrt{\frac{m_{Li}}{2\pi k T_s}} \exp\left(-\frac{m_{Li} v_y^2}{2k T_s}\right), \quad (4)$$

$$f_z(v_z) = \frac{m_{Li}}{k T_s} v_z \exp\left(-\frac{m_{Li} v_z^2}{2k T_s}\right). \quad (5)$$

To reduce computational cost, we do not compute the position of Li atoms at each time step. Instead, we assume that all evaporated neutral lithium is ionized in the plasma, and that the ionization position d_i of particle i follows an $\exp(-d_i/\Lambda_{iz})$ distribution,²⁸ where Λ_{iz} is the three-dimensional electron-induced ionization mean free path given by

$$\Lambda_{iz} = \frac{v_{Li}^s}{\langle \sigma_{iz} v_e \rangle n_e}, \quad (6)$$

where v_{Li}^s is the thermal velocity of the emitted Li atoms based on the surface temperature such that $v_{Li}^s = \sqrt{kT_s/m_{Li}}$ and $\langle \sigma_{iz} v_e \rangle$ is the electron-induced ionization rate taken from Ref. 37 and v_e is the electron thermal velocity.

Based on the cross section coefficients provided in Refs. 18 and 38, we estimate that the mean free paths for elastic collisions, recombination, and most charge-exchange processes exceed the thicknesses of both the sheath and the magnetic presheath. These thicknesses scale with the Debye length and the D^+ Larmor radius, respectively. Among these processes, only $Li + Li^+ \rightarrow Li^+ + Li$ charge-exchange reaction could occur in a very small proportion, within the region of interest. However, for simplicity and to ensure broader applicability of our results, we neglect this specific atomic reaction.

Practically, at each time step, the number of emitted Li atoms is determined from Eq. (3). Each atom has its velocity components

sampled according to Eqs. (4) and (5). The ionization process is assumed not to affect the atomic velocity, so that the components are preserved upon ionization. The ionization position d_i is then sampled from the aforementioned exponential distribution. A Li^+ ion is injected into the corresponding cell provided that an electron with energy $E > I_1$ (with $I_1 = 5.39$ eV the first ionization potential of lithium) is present. If no such electron is available, another electron is sought in neighboring cells. Once a sufficiently energetic electron is identified, the remaining energy $E - I_1$ is evenly shared between the scattered and ejected electrons, and their velocity directions are updated following Ref. 39.

Only Li^+ ions are considered in this study, as the ionization potentials for the second and third ionizations (75.64 and 122.45 eV, respectively) are too high to produce a significant amount of these species.

C. Plasma parameters

The range of nominal plasma densities evaluated spans from 5×10^{18} to 10^{20} m^{-3} , with electron and deuterium temperatures ($kT_e = kT_D$) set to either 50 eV or 25 eV. While these conditions are higher than those typically encountered in classical reactor scenarios—such as the partially detached divertor plasma of ITER¹—they were chosen to reduce computational time while preserving physical relevance.

We also consider magnetic field inclinations ranging from 2.5° to 90° with respect to the wall, since phenomena such as meniscus formation around objects like antennas, undulations due to surface instabilities, and the chaotic magnetic configurations in the edge regions of stellarators⁴⁰ can result in a wide range of inclinations. Regarding magnetic field intensity, we explore values between 1 and 10 T, enabling the study of experimental machines as well as large-scale (low-field) and compact (high-field) reactors.

We also vary the ratio between the incoming lithium and deuterium fluxes, $\gamma_s = \Gamma_{\text{Li}}/\Gamma_D$, by adjusting the wall temperature and the main plasma conditions, in order to capture the effect of the lithium influx on sheath physics and redeposition processes.

The 1D approach adopted here neglects drift effects and spatial parameter variations. As such, it cannot be directly applied to specific configurations like stellarators or tokamaks. Instead, it serves as an exploratory study of the underlying physics. While our results on sheath properties and in-sheath redeposition are general, the uncovered physics may be relevant to other plasma scenarios. Moreover, ones can imagine that conditions explored here could arise in specific future designs, particularly in compact reactors subject to high transient heat loads (depending on their duration),⁴¹ or in low-recycling⁴² and attached regimes.⁴¹

III. REDEPOSITION RESULTS FOR VARIOUS PLASMA AND LIQUID WALL PARAMETERS

A. Sheath build-up, transients, and surface temperature effects

In this section, we consider a deuterium plasma source with $\Gamma_D = 10^{23} \text{ m}^{-2} \text{ s}^{-1}$ and ion and electron temperatures $kT_D = kT_e = 50$ eV. A magnetic field of $B = 2.5$ T is inclined at 30° with respect to the liquid metal surface. The lithium flux at the wall, Γ_{Li} , is varied across simulations by adjusting the surface temperature T_s . As T_s increases, Li atoms evaporate with higher initial energy, resulting in a

slightly longer ionization mean free path Λ_{iz} and deeper injection of Li^+ ions into the plasma [see Eq. (6)]. Li^+ ions are tagged according to their emitting wall. If an ion redeposits on the same wall from which it originated, it contributes to the redeposited flux Γ_{red} , such that $\Gamma_{\text{red}} = \gamma_{\text{red}} \Gamma_{\text{Li}}$. This relation expresses that the redeposited flux equals the total evaporated flux, given by Eq. (3), multiplied by the sheath redeposition fraction γ_{red} .

The time required for the system to reach steady-state conditions, i.e., temporal convergence of γ_{red} [Fig. 3(b)], strongly depends on the ratio between the Li emitting flux and the plasma source one $\gamma_s = \Gamma_{\text{Li}}/\Gamma_D$ and the final state of the plasma-wall transition. For low $\gamma_s = 0.1$, steady state is achieved after about $15 D^+$ cyclotron periods, as shown in Fig. 2(a), while the average electron temperature remains nearly constant over time, close to the nominal value, as seen in Fig. 3(a). In this case, the presence of Li has no major effects on the potential profile, with a potential drop between the wall and the plasma center of approximately $3.4kT_e$. This includes the expected drop between the entrance of the magnetic pre-sheath and the wall given by the well-known expression,⁴³

$$\frac{e\phi_s}{kT_e} = \frac{1}{2} \ln \left[\frac{2\pi m_e}{m_i} \left(1 + \frac{T_i}{T_e} \right) \right], \quad (7)$$

where m_i is the ion mass and T_i is the ion temperature. The latter potential drop is about $2.8kT_e$ in absolute value for pure D^+ and is combined with the potential drop in the classical collisional or inertial pre-sheath,^{44,45} which in total gives a bit more than $3kT_e$.

For $\gamma_s = 1$, the stabilization time for the sheath and pre-sheath remains close to 15 ion gyroperiods, as shown in Fig. 2(b). The electron temperature slightly decreases over the first ten periods before stabilizing near the nominal temperature at 48 eV, as evidenced by the green curve of Fig. 3(a). This delay occurs because colder electrons, produced from Li ionization, take time to diffuse through the plasma volume and reinitialize at the Maxwellian temperature. In this case, the potential drop between the plasma center and the wall remains unchanged ($\simeq 3.4kT_e$). However, the spatial variation of the plasma potential ϕ is significantly altered, flattening considerably between the point marked $\langle d_{iz} \rangle$ in Fig. 2 and the plasma center. Here, $\langle d_{iz} \rangle$ represents the projection of the 3D mean free path Λ_{iz} along the z-axis (normal to the surface), while the shaded region in Fig. 2 indicates where 25%–95% of Li^+ ions are injected. This flattening results from density accumulation in the injection region, such that the electron density n_e at $\langle d_{iz} \rangle$ becomes comparable to the one at the plasma center [Fig. 2(e)].

For $\gamma_s = 10$, a transient bump in the plasma potential appears around 15 ion gyroperiods in the injection zone, which tends to vanish in steady-state [Fig. 2(c)]. This bump is caused by a local density increase: Li^+ ions, being slower than D^+ ions and electrons, accumulate in the injection region and must be accelerated both toward the wall and into the plasma to maintain ambipolarity until steady state is reached. Such a potential bump was previously demonstrated by Benilov and Benilova,⁴⁶ where ions of the same species as the plasma were injected into the sheath to ensure ambipolarity in the steady state. Furthermore, for this high Li^+ flux, a sharp drop in electron temperature by 7 eV is observed at the beginning of the simulation [Fig. 3(a)], before it gradually recovers after about ten periods. In steady state, the plasma potential is relatively flat, as in the previous case, but the total potential drop ($\simeq 4.2kT_e$) is linked to Li being the dominant species in

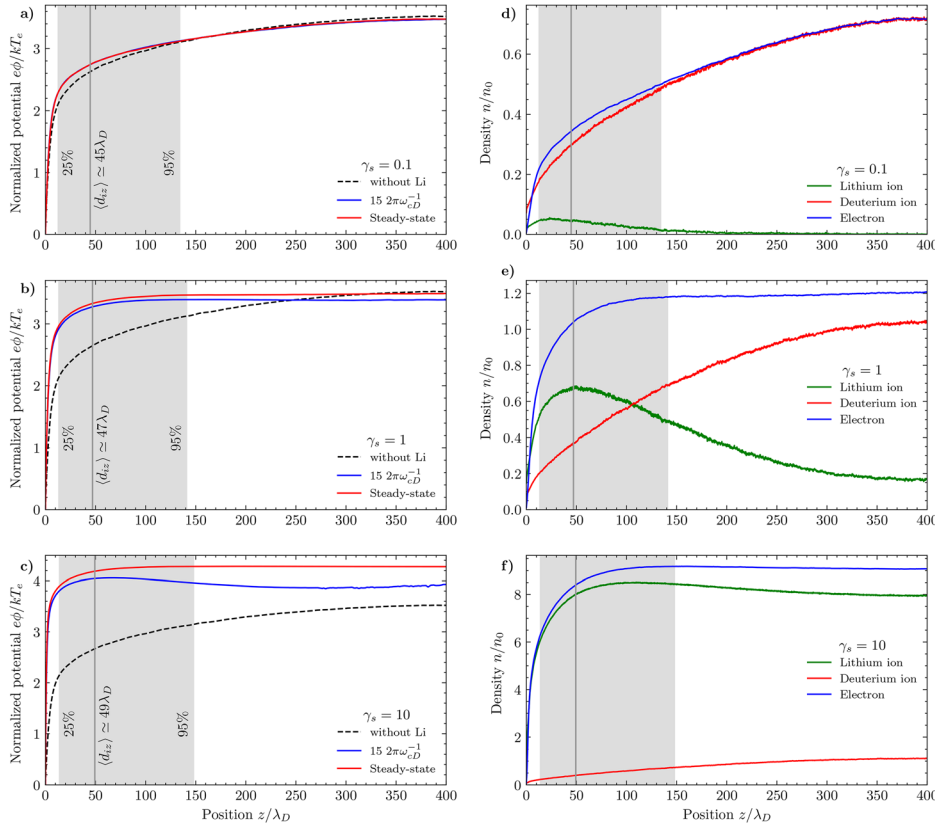


FIG. 2. (a)–(c) potential and (d)–(f) steady-state charged particle density ($n_0 = 10^{19} \text{ m}^{-3}$) steady-state profiles for various flux ratios ($\gamma_s = \Gamma_{\text{Li}}/\Gamma_{\text{D}}$) obtained by surface temperature variations. Cases without Li reach the steady-state (~ 10 periods) much faster than ones with Li (~ 100 periods). $\langle d_{iz} \rangle$ represents the mean free path projected along the z -axis, normal to the surface, while the shaded region indicates where 25%–95% of Li^+ ions are injected.

the plasma [Fig. 2(c)]. Using Eq. (7) and assuming the lithium temperature to be close to the wall temperature T_s , with m_i denoting the lithium ion mass, the expected potential drop across both the magnetic pre-sheath and the Debye sheath is approximately $3.8\kappa T_e$. When adding the pre-sheath potential drop ($\simeq 0.5\kappa T_e$), the result is in very good agreement with our simulation outcomes. Finally, the spatial variation of the electric potential in the injection zone, which depends on the ratio $\gamma_s = \Gamma_{\text{Li}}/\Gamma_{\text{D}}$ as previously discussed, has a strong impact on redeposition. For $\gamma_s = 0.1$, lithium can be regarded as a trace impurity since it does not alter the sheath structure, which generates a sufficient electric force to pull nearly all lithium ions back to the wall [redeposition rate $\simeq 100\%$, see Fig. 3(b)]. In contrast, for $\gamma_s = 10$, lithium becomes the dominant plasma species and the redeposition rate decreases to 96.5% due to the flattening of the potential, which weakens the electric force and thus reduces redeposition. The simulated surface temperatures and the corresponding results are summarized in Table I for different magnetic field orientations, which will be discussed in the following Sec. III B.

B. Influence of magnetic field inclination

The electric potential profiles computed for several magnetic field inclinations are shown in Fig. 4(a), alongside the corresponding D^+ and Li^+ densities shown in Fig. 5(a) for $\gamma_s = 1$ (the incoming Li^+ flux equals that of D^+), $B = 2.5 \text{ T}$, and $\kappa T_e = \kappa T_D = 50 \text{ eV}$. For comparison, the pure deuterium plasma case is shown in Fig. 4(a). For all

magnetic field inclinations considered, the potential drop between the 1D mean free path $\langle d_{iz} \rangle$ and the wall increases in the presence of the intense Li^+ source [Fig. 4(a)]. This behavior is attributed to the sheath contraction near the wall due to the significant increase in local plasma density, in excess of the pure D case, because of Li^+ accumulation.

Moreover, the flattening of the potential between the 1D mean free path $\langle d_{iz} \rangle$ and the center of the simulated system, as evidenced in Sec. III A for $\theta = 30^\circ$, is not observed at grazing incidences [e.g., $\theta = 5^\circ$, in Fig. 4(a)]. This behavior results from the increase in D^+ density n_D at the plasma center for such low field orientations, which shapes and rounds the potential profile, since n_D exceeds the Li^+ density peak located around $\langle d_{iz} \rangle$ [see Fig. 5(a)].

This increase in D^+ density with decreasing θ arises from the system dynamics. Since D^+ ions follow the field lines throughout most of the quasi-neutral plasma, they experience a reduction in velocity along the z -direction as θ decreases. The deuterium flux along z can be written as $\Gamma_D(z) = n_D v_D \sin \theta$, where v_D is a characteristic velocity along the field line. Thus, in order to maintain the constant imposed Γ_D , the D^+ density n_D must increase as θ decreases. Conversely, lithium accumulates near the wall, with its density varying only weakly with θ : in this region the strong electric field accelerates Li^+ ions toward the wall, deviating their trajectories from the B -field lines—unlike the D^+ ions in the plasma center, as discussed above. This balance between the two ion species explains the angular dependence of the potential profile, ranging from a flattened shape at large incidence angles to a more rounded one at grazing magnetic field orientations, Fig. 4.

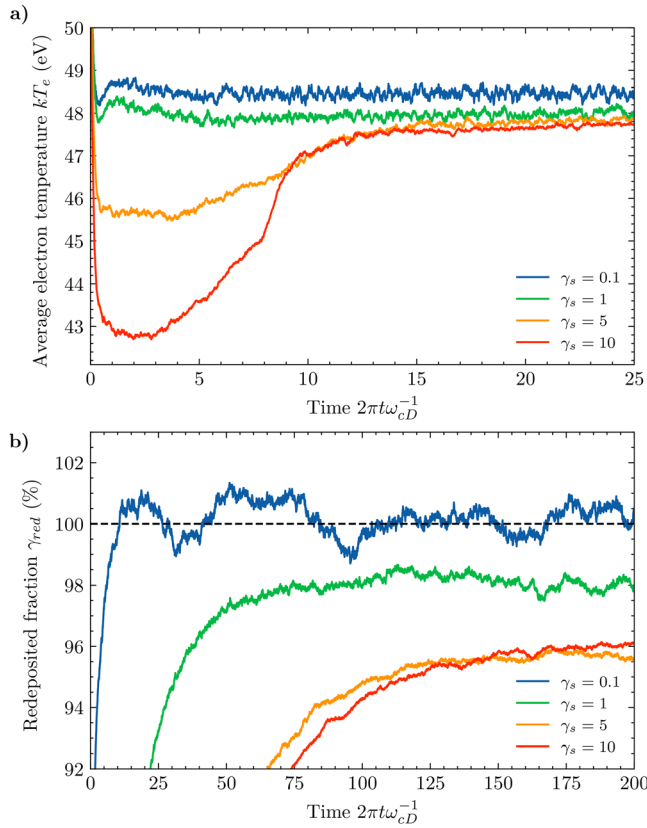


FIG. 3. (a) Time evolution of the average electron temperature and (b) of the redeposited fraction $\gamma_{red}(30^\circ)$ for various flux ratios, obtained through surface temperature variations: Low flux ratios ($\gamma_s = 0.1$ and 1) exhibit rapid convergence, while higher values ($\gamma_s = 5$ and 10) require over 200 periods to fully converge. Interestingly, the blue curve gives in average a redeposited fraction close to 100%. Observed high-amplitude oscillations for low flux ratio value, in particular $\gamma_s = 0.1$, are due to the low number of Li ions compared to other cases where the evaporation flux is higher.

This angular evolution of the potential profile strongly impacts the Li^+ redeposition rate γ_{red} , as shown in Fig. 5(b) for $\gamma_s = 1$ and in Table I for other wall temperatures. The flattening of the potential lowers the electric field in the Li ionization region and thereby reduces redeposition. At more grazing incidences, however, the potential profile tends to become more evanescent, which enhances redeposition

and can lead to fractions γ_{red} exceeding 99% for small angles, even at high γ_s (higher surface temperature).

Interestingly, the potential bump observed in the transients for $\theta = 30^\circ$ (see Sec. III A) can persist into steady states if the Li^+ injection is sufficiently strong to locally enhance the plasma density compared to the plasma bulk [Fig. 4(b)]. The emergence of this potential bump in the Li^+ injection region, which peaks at $\theta = 60^\circ$ and diminishes for both higher and lower inclinations, explains the angular variation of γ_{red} , notably its minimum around 60° , as shown in Fig. 5(b). Analysis of the ionization position distributions of redeposited Li^+ ions reveals that the redeposition fraction is correlated with the spatial distribution of ionized lithium atoms. Notably, nearly all lithium atoms ionized within $125 \lambda_D$ from the wall (approximately 93% of the total Li^+ population) are redeposited. The redeposition of the remaining few percent depends on the potential gradient in the pre-sheath and on the potential bump. Despite its relatively small amplitude (< 1 eV), the bump is sufficient to prevent cold Li^+ ions from redepositing and explains the redeposition rate trend evidenced with respect to the magnetic field inclination.

C. Influence of plasma density

The ionization mean free path Λ_{iz} , or its 1D projection $\langle d_{iz} \rangle$, strongly affects the redeposition rate, as it determines the plasma region (sheath or pre-sheath) where impurities first become sensitive to the electromagnetic field and can eventually be pulled back to the wall. As explained in Sec. II, the main deuterium plasma is injected into the simulated system at a constant flux Γ_D , chosen to be of the order of the Bohm flux. Since the ionization mean free path depends on both plasma density and temperature [see Eq. (6)], varying Γ_D at constant plasma temperature ($T_e = T_D$) allows one to tune the plasma density, denoted n_{nom} . For this set of simulations, the wall temperature T_s was kept the same as in Sec. III A, such that $\Gamma_{Li} = 10^{23} \text{ m}^{-2} \text{ s}^{-1}$ with $B = 2.5 \text{ T}$ and $\theta = 30^\circ$. It should be noted, however, that fixing T_s introduces the limitation of altering the ratio $\gamma_s = \Gamma_{Li}/\Gamma_D$ between simulations. As a result, the effect of varying $\langle d_{iz} \rangle$ on plasma structure and redeposition is influenced by the relative contribution of Li compared to D in the plasma (see Sec. III A).

Table II summarizes the corresponding 1D-averaged ionization length $\langle d_{iz} \rangle$ for the different plasma densities, along with the values of γ_s and the redeposition rate γ_{red} obtained. As the average injection point $\langle d_{iz} \rangle$ of Li^+ in the plasma increases (from $14.6 \lambda_D$ to $65.6 \lambda_D$), the redeposition rate decreases from 100% to 95%. This trend is consistent with the electric potential profiles shown in Fig. 6(a) for the various plasma densities [corresponding Li^+ density profiles are depicted in

TABLE I. Lithium flux Γ_{Li} , flux ratio γ_s , 1D-projection of the ionization mean free path $\langle d_{iz} \rangle$ and redeposited fraction γ_{red} as a function of the surface temperature T_s , for three different magnetic inclinations $\theta = 5^\circ, 15^\circ$, and 30° . Uncertainties corresponds to the standard deviations of γ_{red} related to the used averaging window.

$T_s(^{\circ}\text{C})$	$\Gamma_{Li}(\text{m}^{-2} \text{ s}^{-1})$	γ_s	$\langle d_{iz} \rangle / \lambda_D$	$\gamma_{red}(5^\circ)(\%)$	$\gamma_{red}(15^\circ)(\%)$	$\gamma_{red}(30^\circ)(\%)$
487.22	1×10^{22}	0.1	44.0	99.5 ± 0.8	99.8 ± 1.2	99.8 ± 0.8
541.74	5×10^{22}	0.5	45.7	...	...	99.8 ± 0.3
567.77	1×10^{23}	1	46.9	100.0 ± 0.3	100.0 ± 0.2	98.3 ± 0.2
604.92	2.5×10^{23}	2.5	47.0	100.0 ± 0.1	98.3 ± 0.1	96.4 ± 0.2
635.33	5×10^{23}	5	47.6	99.4 ± 0.1	97.4 ± 0.1	95.9 ± 0.1
667.97	1×10^{24}	10	49.3	...	...	96.5 ± 0.1

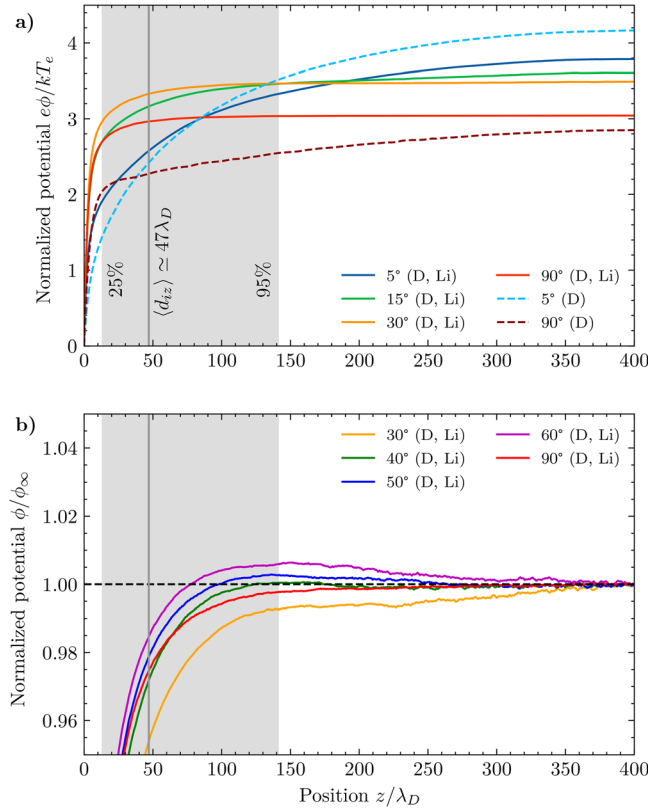


FIG. 4. (a) Normalized potential $e\phi/kT_e$ profiles for several magnetic inclinations: Dashed lines correspond to simulations with only D^+ ions, while solid lines correspond to simulations with both Li^+ and D^+ ions. (b) Evolution of the potential ϕ/ϕ_∞ near the ionization region with the magnetic inclination: ϕ_∞ corresponds to the potential in the simulated plasma center, i.e., at $400\lambda_D$.

Fig. 6(b)]. At high plasma densities, Li ions are injected closer to the wall ($z < 20\lambda_D$), within the Debye sheath where the strong electric field drives high redeposition. In these cases, only a small fraction of Li remains in the edge plasma [see Fig. 6(b)]. Conversely, as $\langle d_{iz} \rangle$ (the 1D ionization mean free path) increases, a larger fraction of impurities penetrates toward the plasma center, since Li^+ ions are injected into regions with weaker electric fields.

D. Influence of magnetic field strength

Assessing the redeposition of Li emitted from a liquid PFC under stronger magnetic fields is particularly relevant for high-field reactor designs, targeting $B \simeq 10$ T with high-temperature superconductors (HTS). However, such simulations are computationally demanding for our PIC code, since both the cell size and time step are constrained by the electron Larmor radius and cyclotron period, which decrease with B . This results in a finer computational grid and an increased number of superparticles and time steps to reach steady state. For this reason, we restricted the study to a limited set of simulations with $\gamma_s = 1$ ($\Gamma_D = \Gamma_{\text{Li}}$), $\Gamma_D = 10^{23} \text{ m}^{-2} \text{ s}^{-1}$, $kT_D = kT_e = 50$ eV, and a magnetic-field incidence of $\theta = 30^\circ$, while varying B from 1 to 10 T. In fact, ion motion in the sheath and pre-sheath regions is governed by

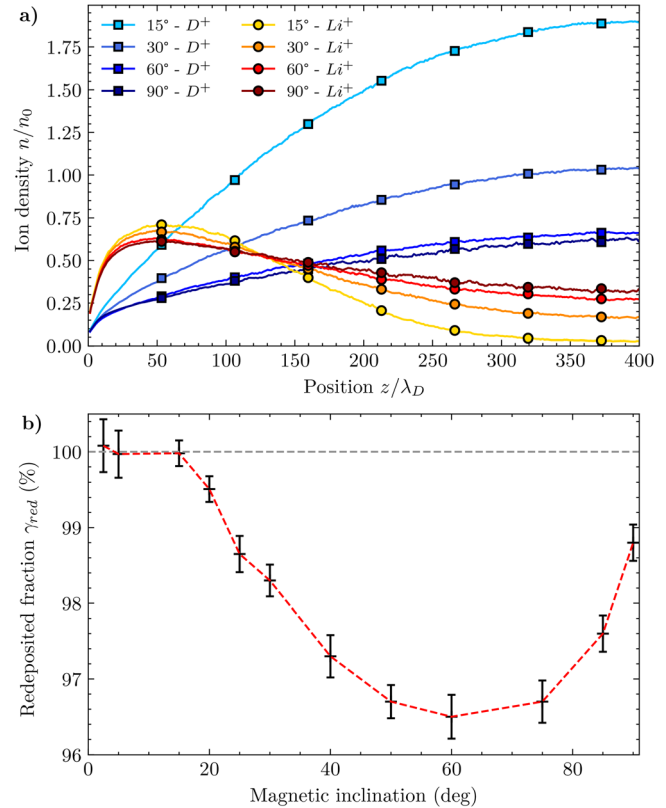


FIG. 5. (a) Li^+ and D^+ density ($n_0 = 10^{19} \text{ m}^{-3}$) profile for several magnetic inclinations. (b) Evolution of the redeposited fraction γ_{red} with the magnetic inclination for the $\gamma_s = 1$ configuration presented previously.

TABLE II. Flux ratio γ_s , 1D-projection of the ionization mean free path $\langle d_{iz} \rangle$ and redeposited fraction γ_{red} at $\theta = 30^\circ$ as a function of the nominal plasma density n_{nom} for a surface temperature T_s equals to 568°C and a plasma energy $kT_e = kT_D = 50$ eV.

$n_{\text{nom}}(\text{m}^{-3})$	γ_s	$\langle d_{iz} \rangle/\lambda_D$	$\gamma_{\text{red}}(30^\circ)(\%)$
5×10^{18}	2	65.6	95.0 ± 0.3
1×10^{19}	1	46.9	98.3 ± 0.2
5×10^{19}	0.2	20.9	99.8 ± 0.7
1×10^{20}	0.1	14.6	100.2 ± 1.1

the competition between electric and magnetic fields. While the electric field E accelerates ions toward the wall, the magnetic field B constrains their trajectories along the field lines. As a result, the redeposition rate strongly depends on the E/B ratio, as shown in Fig. 7, with γ_{red} decreasing significantly with B , from almost 99% at 1 T down to 96% at 10 T.

As the magnetic field strength increases, the transit time of ions from their injection point in the plasma to the wall becomes longer [Fig. 8(b)], due to the E - B competition mentioned above. This constrained motion leads to a stronger accumulation of Li^+ in the sheath and pre-sheath (when the mean free path is comparable to their

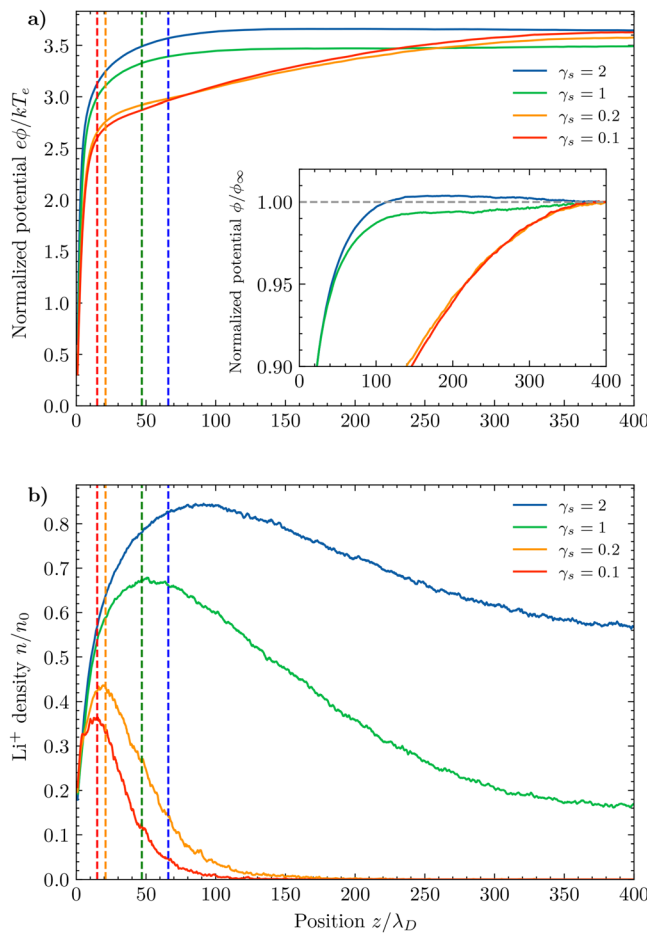


FIG. 6. (a) Normalized potential and (b) Li^+ density ($n_0 = 10^{19} \text{ m}^{-3}$) profiles for several flux ratios γ_s obtained by nominal deuterium plasma density variations at fixed surface temperature $T_s = 568^\circ\text{C}$. The vertical dashed lines indicate $\langle d_L \rangle$ for the different simulations, with the same color legend applied.

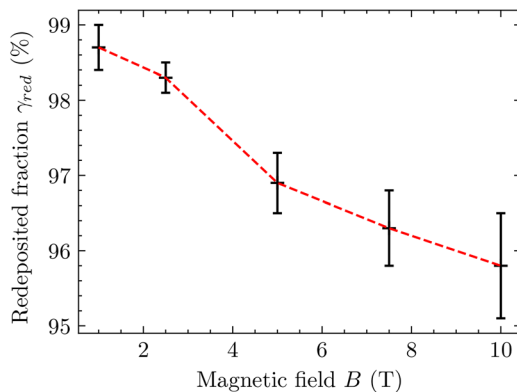


FIG. 7. Redepleted fraction γ_{red} at $\theta = 30^\circ$ for several magnetic intensities B in the $\gamma_s = 1$ configuration where $T_s = 568^\circ\text{C}$.

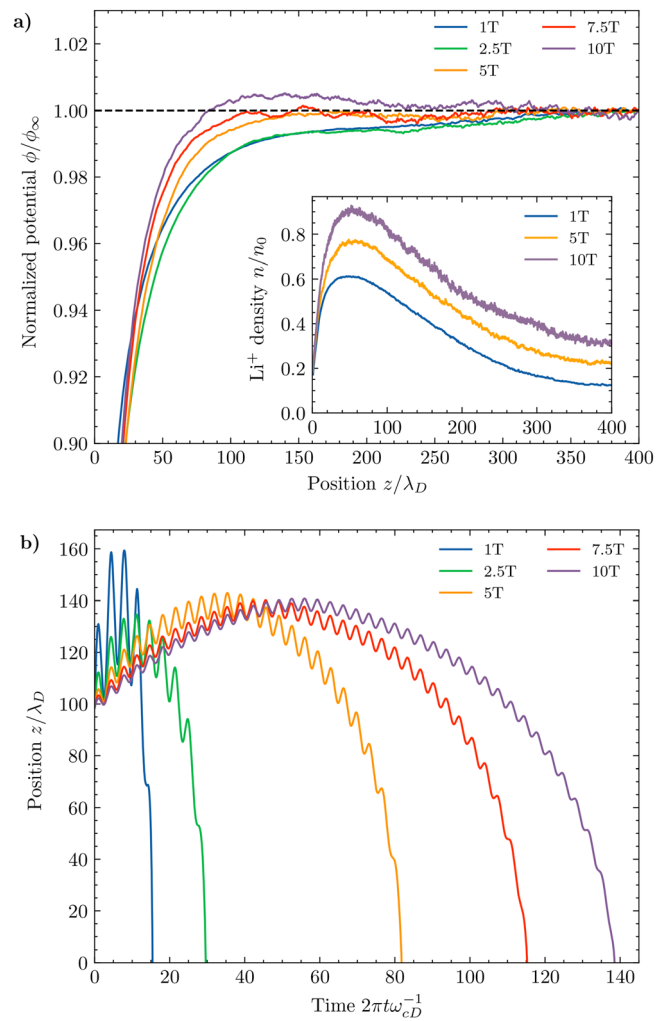


FIG. 8. (a) Normalized potential ϕ/ϕ_∞ and lithium ion density ($n_0 = 10^{19} \text{ m}^{-3}$) profiles for several magnetic intensities. (b) Trajectories of Li^+ ion with an initial velocity equal to $10 v_{Li}^s$ launched in the 5 T electric field configuration as a function of the magnetic intensity.

characteristic lengths, as in our simulations). Consequently, the Li^+ density increases near the wall [see inset in Fig. 8(a)], which slightly flattens the potential toward the plasma center (as discussed in Sec. III B), and may even produce a potential bump in the injection region [Fig. 8(a), for $B = 10$ T]. Both of these effects on the potential profile contribute to a significant reduction in the redeposition rate for strongly magnetized sheaths.

IV. CONCLUSIONS

This study was motivated by recent developments in compact nuclear fusion reactors, which call for new plasma-facing component (PFC) solutions—and in particular, liquid metal walls—to withstand the higher heat fluxes specific to these machines. Our investigation focused on lithium redeposition on plasma-facing walls in magnetized deuterium plasma sheaths, using an in-house 1D3V Particle-in-Cell

code. The chosen plasma input parameters are intended to explore reactor regimes that are more compact than large-scale experiments such as ITER. The code self-consistently describes the impact of lithium evaporation on the sheath structure, the electrostatic potential, and consequently the redeposition process. This work should be regarded as a preliminary yet relevant step toward addressing the underlying physical problem, although many fundamental processes are not yet included. In particular, neutral particles, several key atomic processes, and the coupling between plasma and surface remain to be incorporated in future studies.

Contrary to heavy impurities such as tungsten, which are mainly produced by sputtering on solid PFCs, lithium exhibits a much higher erosion rate due to its strong evaporation contribution. Lithium must therefore be treated not as a trace impurity but as a full-fledged plasma species that contributes to the electrostatic balance, leading to a multi-ion sheath in which it competes with the main plasma species (deuterium in the present case). Within the framework of the present model and assumptions, we have shown that lithium redeposition is governed by four key parameters: (1) the ratio γ_s between the outflux of evaporated and ionized Li^+ and the influx of D^+ , (2) the magnetic field inclination θ with respect to the wall, (3) the ionization mean free path Λ_{iz} , and (4) the magnetic field intensity B . Li redeposition is generally high, exceeding 95%, particularly for low γ_s and for grazing incidence of the magnetic field. However, higher values of the four parameters tend to reduce the redeposited fraction γ_{red} , except for the magnetic inclination θ , where γ_{red} angular variations can be non-monotonic. Results can be interpreted as a function of plasma parameters (density and energy). In this case, for a given surface temperature, a decrease in the plasma temperature or density leads to an increase in both γ_s and Λ_{iz} , impacting redeposition and sheath properties accordingly. Also, when the incoming Li^+ flux is comparable to or larger than the main plasma flux, lithium accumulates at a distance from the wall of about the one-dimensional ionization mean free path (in the sheath and pre-sheath regions for the simulated plasma parameters). This accumulation alters the expected plasma-wall transition structure, leading to a thinner sheath compared to the pure deuterium plasma case. It can also give rise to a bump in the electrostatic potential profile, which prevents a fraction of the lithium ions from redepositing on the emitting wall.

Although this study highlights the impact of strong lithium evaporation on sheath physics and its implications for the redeposition process, and consequently SOL boundary conditions, it cannot provide a definitive answer regarding the suitability of liquid lithium as a plasma-facing material. Indeed, boundary conditions are also strongly influenced by wall pumping properties, neutral fluxes, self-consistent ionization mean free path variations, and other non-linear mechanisms—not considered in this study. Ultimately, modifications to these boundary conditions could lead to significant alterations in core and SOL physics compared to tungsten PFCs. This could potentially transform tokamak and stellarator operation by introducing new degrees of freedom and physical behaviors. Finally, the question of sputtered lithium redeposition must be investigated, as its mean free path may lead to a high number of ionizations outside the sheath and pre-sheath regions.

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AUTHOR DECLARATIONS

Conflict of Interest

Yes, the authors declare the following financial interest/personal relationships which may be considered as potential competing interest: Francesco Volpe has patents #WO2023194373A1, #WO2023194368A1, and #WO2023194366A1 issued to Renaissance Fusion. The other authors declare that they have no known competing financial interest or personal relationships that could have influenced the work reported in this paper.

Author Contributions

R. Avril: Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (equal); Software (equal); Writing – original draft (lead). **S. Heuraux:** Formal analysis (supporting); Writing – review & editing (equal). **E. Gravier:** Supervision (equal); Writing – review & editing (equal). **F. A. Volpe:** Funding acquisition (equal); Writing – review & editing (equal). **J. Moritz:** Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Software (equal); Supervision (lead); Writing – original draft (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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